ALICE L WALTON SCHOOL OF MEDICINE

Agents Used in the Treatment of Inflammatory Bowel Disease (IBD)

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Learning Objectives (2)

After completing this brick, you will be able to:

1. Compare and contrast the mechanisms of action of drugs used to treat inflammatory bowel disease
2. Analyze differences in mechanisms and adverse effects among inflammatory bowel disease treatment agents

CASE CONNECTION

MJ, a 34-year-old female, presents at a gastroenterology clinic with a two-month history of persistent abdominal pain, bloody diarrhea, and unintentional weight loss. Her symptoms have progressively worsened, and she reports urgency and cramping with each bowel movement. AJ, a second-year medical student on her internal medicine rotation, is asked to consider how drug therapy could address MJ's symptoms and what side effects should be monitored.

What agents are most suitable for MJ, and how might their mechanisms relate to her presentation? What adverse effects should AJ anticipate and monitor? Consider your answers as you read, and we'll revisit MJ at the end of the brick.

GO TO CONCLUSION ↓

Inflammatory bowel disease (IBD) encompasses chronic inflammatory conditions of the gastrointestinal tract, primarily Crohn disease (CD) and ulcerative colitis (UC). The pathophysiology involves dysregulated immune activation, mucosal barrier disruption, and sustained inflammatory signaling. Pharmacological management targets these processes through several distinct drug classes, each acting at different points in the inflammatory cascade. Understanding these mechanisms is essential for selecting appropriate therapy and anticipating clinical outcomes.

Mechanisms of Action of Drugs Used to Treat Inflammatory Bowel Disease and their ADRs (LOs # 1 and 2)

Classification of agents used in the treatment of IBD

Aminosalicylates Mechanism

Aminosalicylates (5-ASAs), such as mesalamine, sulfasalazine, and balsalazide, exert their anti-inflammatory effects primarily through local action on the intestinal mucosa.

Mechanism of action: The active moiety, 5-aminosalicylic acid (5-ASA), inhibits the nuclear factor kappa B (NF-κB) signaling pathway, reducing the transcription of pro-inflammatory cytokines including interleukin-1 (IL-1), tumor necrosis factor-alpha (TNF-α), and interleukin-6 (IL-6). This results in decreased mucosal inflammation and promotes epithelial healing.

5-ASAs (salicylic acid = similar to aspirin's MOA) also inhibit cyclooxygenase (COX) and lipoxygenase enzymes, thereby reducing prostaglandin and leukotriene synthesis—key mediators of intestinal inflammation. Because their action is largely topical, delivery formulations are designed to release the drug at specific sites in the colon or small intestine, depending on the disease location.

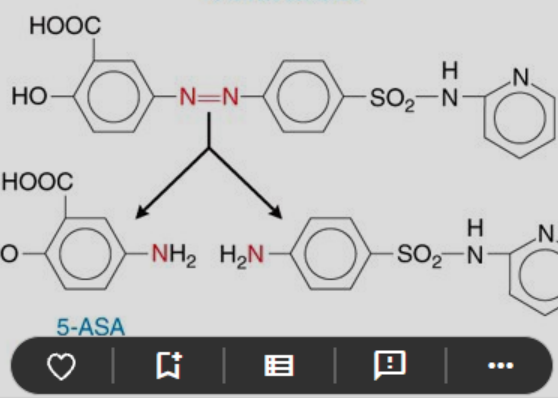


Figure 1: Structure of Sulfasalazine

The *azo* linkage (N=N) in sulfasalazine prevents absorption in the stomach and small intestine, and the individual components are not liberated until colonic bacterial *azoreductases* cleave the bond for local effect (Figure 1). Therefore, to provide localized effects many of the formulations have pH-sensitive coating that help with the release of the drug.

Adverse effects: Aminosalicylates are generally well tolerated. Sulfasalazine, which contains a sulfa moiety, is associated with more side effects than others. It can cause folate deficiency which may require folic acid supplementation. Common ADRs include nausea, headache, and dyspepsia, often related to the sulfonamide component. Hypersensitivity reactions, including rash, have been reported. Monitoring renal function is important since most 5-ASAs are renally excreted.

Corticosteroids Mechanism

Corticosteroids, including prednisone, methylprednisolone, and budesonide, are potent anti-inflammatory agents used in IBD flares.

Mechanism of action: They act by binding to intracellular glucocorticoid receptors (GRs), and the resulting GR-ligand complex translocates to the nucleus, where it modulates gene transcription. This translocation suppresses the expression of numerous pro-inflammatory mediators, including cytokines, chemokines, and adhesion molecules.

Corticosteroids also stabilize cell membranes, reduce vascular permeability, and inhibit phosphol acid release and downstream prostaglandin and leukotriene production.

Budesonide is notable for its high first-pass hepatic metabolism (CYP3A4), which limits systemic exposure and reduces the risk of systemic adverse effects compared to conventional corticosteroids.

Adverse drug effects: The significant adverse events associated with conventional glucocorticoids such as prednisone limit their long-term use. These are numerous, but among the more common are skin and soft-tissue manifestations, including skin thinning and the development of Cushingoid features (weight redistribution and weight gain). Other side effects include cardiovascular events as well as psychiatric and cognitive effects.

Despite their efficacy in acute flares, corticosteroids are not recommended for long-term maintenance therapy due to their significant side effect profile.

Immunomodulators

Immunomodulators used in IBD include thiopurines (azathioprine and 6-mercaptopurine) and methotrexate (MTX).

Mechanism of action: Thiopurines are prodrugs converted to active metabolites—most importantly 6-thioguanine nucleotides (6-TGN)—which incorporate into DNA and RNA, inhibiting purine synthesis and inducing apoptosis of activated T lymphocytes. This reduces the proliferation of immune cells driving intestinal inflammation.

CLINICAL CORRELATION

Thiopurines are prin d as steroid-sparing agents.

Methotrexate inhibits dihydrofolate reductase (DHFR), an enzyme essential for folate metabolism and nucleotide synthesis. By depleting folate, MTX impairs rapidly dividing immune cells and reduces pro-inflammatory cytokine production (Figure 2).

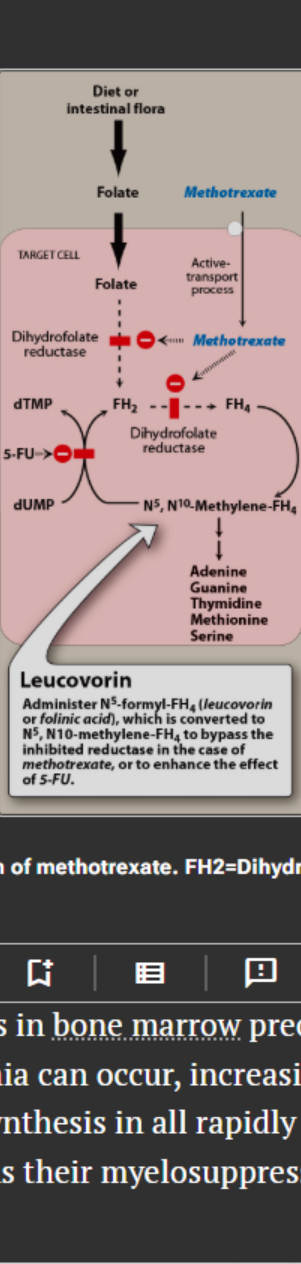
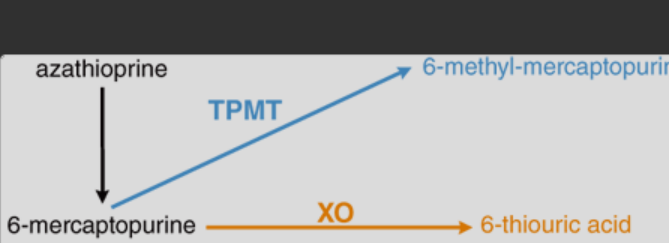


Figure 2: Mechanism of action of methotrexate. FH2=Dihydrofolate, FH4= tetrahydrofolate

Adverse effects: Thio on due to their inhibition of purine synthesis in bone marrow precursors. Leukopenia, thrombocytopenia, and anemia can occur, increasing infection risk. Thiopurines inhibit purine synthesis in all rapidly dividing cells—not just immune cells—which explains their myelosuppressive effects on bone marrow.



Source: Laurence L. Brunton, Björn C. Knollmann: *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e*: Copyright © McGraw Hill. All rights reserved.

Figure 3: Metabolism of Azathioprine. (Thiopurine methyltransferase (TPMT)

Thiopurine methyltransferase (TPMT) enzyme activity varies genetically; patients with low TPMT activity accumulate toxic 6-thioguanine nucleotides (6-TGN) levels, leading to dose-dependent bone marrow suppression (myelosuppression). Thus, TPMT testing before initiation is essential. Hepatotoxicity and pancreatitis are additional concerns, particularly early in therapy.

Xanthine oxidase (XO) in the small intestine and liver converts mercaptopurine to thiouric acid, which has no therapeutic activity. Allopurinol, a medication used in gout treatment inhibits XO. Inhibition of XO by allopurinol diverts 6-mercaptopurine to more active metabolites, such as 6-thioguanine, increasi ntial toxic effects. Therefore, the use of allopurinol with azathioprine should be avoided.

Methotrexate's adverse effects include nausea, mucositis, hepatotoxicity, and pulmonary toxicity (pneumonitis). Methotrexate's inhibition of DHFR depletes folate in rapidly dividing mucosal epithelial cells, causing mucositis and gastrointestinal toxicity. Folic acid supplementation is often used to reduce these incidences.

What are the three main drug classes used in IBD and their primary mechanisms?

Biologics used in IBD treatment

Cytokines are central orchestrators of intestinal inflammation in IBD. TNF-α, IL-6, IL-12, and IL-23 drive the activation and differentiation of T helper cells, perpetuating mucosal injury.

Anti–TNF-α

Mechanism of action: These drugs bind to and neutralize both soluble and membrane-bound TNF-α, one of the principal cytokines mediating the TH1 immune response characteristic of Crohn disease. This interrupts downstream NF-κB activation and reduces the production of additional pro-inflammatory mediators, leading to mucosal healing (Figure 4).

Infliximab, adalimumab, certolizumab pegol, and golimumab are monoclonal immunoglobulins that have been developed for the treatment of chronic inflammatory diseases. Infliximab is a chimeric antibody (25% mouse, 75% human), whereas adalimumab and golimumab are fully humanized antibodies. *Certolizumab pegol* is a humanized fragment antigen binding that is "pegylated" (i.e., bound to a polyethylene glycol polymer to increase serum half-life of the parent compound).

Adverse effects: Biologic agents, particularly anti-TNF therapies, increase the risk of serious infections (including reactivation of latent tuberculosis), infusion reactions, and, rarely, demyelinating disorders or lymphoma. The primary adverse effect of anti-TNF agents is infection resulting from suppression of the inflammatory response. Screening for latent TB and hepatitis B is mandatory before initiating biologics. Anti-TNF agents, by neutralizing TNF-α, impair granuloma formation, which is essential for containing mycobacterial infections—explaining the risk of latent TB reactivation.

Both acute (fever, chills, urticaria, or even anaphylaxis) and subacute (serum sickness–like) are common reactions with infliximab. This is attributed to the chimeric nature of infliximab.

IL-12/23 and Jak inh

Mechanism of action: Ustekinumab targets the shared p40 subunit of IL-12 and IL-23, inhibiting Th1 and Th17 cell differentiation—pathways particularly implicated in CD. Small molecule inhibitors such as tofacitinib (-*nib*=small molecule tyrosine kinase inhibitor) block Janus kinase (JAK) signaling, which is required for cytokine receptor signal transduction. By interrupting JAK-STAT pathways, tofacitinib broadly suppresses cytokine-driven inflammation (Figure 4).

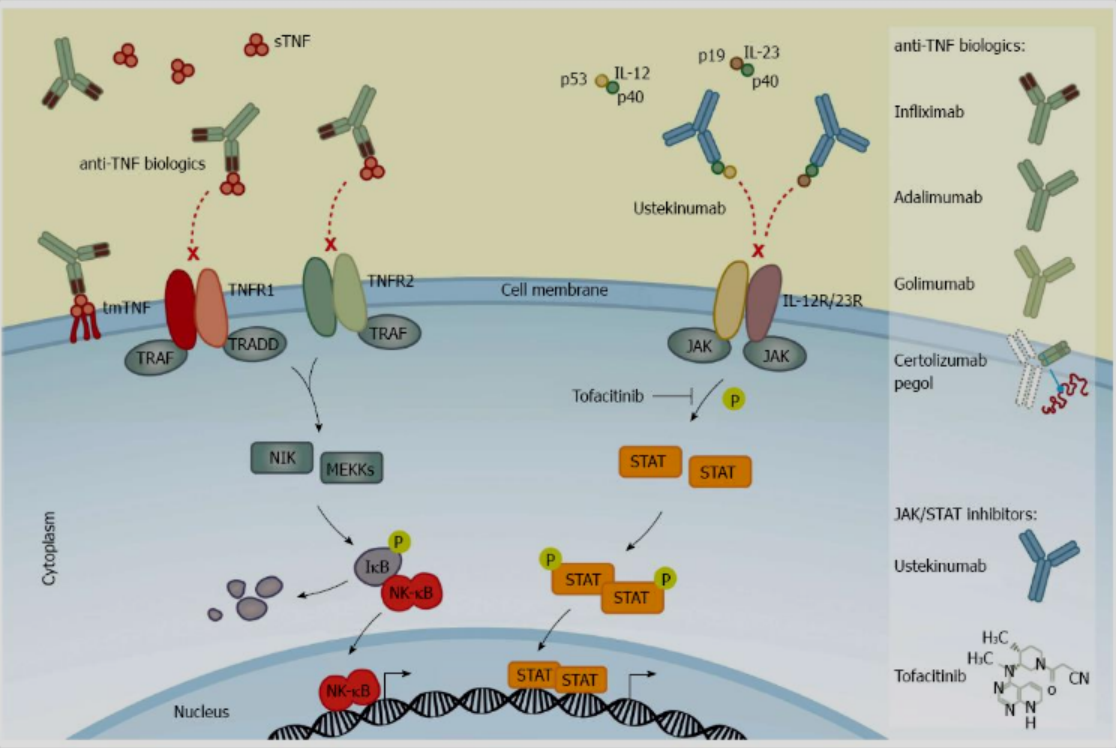


Figure 4: Mechanism of action of biologics used in IBD treatment

Leukocyte Trafficking

Leukocyte migration from the bloodstream into inflamed intestinal tissue is a key amplifier of IBD pathology. This process depends on adhesion molecules and integrins expressed on immune cells.

Anti-integrin biologics

Mechanism of action: Vedolizumab, an anti-integrin biologic, selectively blocks the α4β7 integrin on lymphocytes, preventing their binding to mucosal addressin cell adhesion molecule-1 (MAdCAM-1) on gut endothelium. This gut-selective mechanism reduces lymphocyte infiltration into the intestinal mucosa without broadly suppressing systemic immunity.

Natalizumab is a chimeric (human-mouse) monoclonal antibody against $\alpha_4\beta_1$ and $\alpha_4\beta_7$ integrins on leukocytes involved in inflammation. The drug must not be used in patients with immunodeficiency or in immunosuppressed patients due to a possible increased risk of progressive multifocal leukoencephalopathy, an often lethal viral infection (human polyomavirus 2).

CLINICAL CORRELATION

Progressive multifocal leukoencephalopathy (PML) is a rare, often fatal demyelinating disease of the central nervous system caused by reactivation of the JC virus (John Cunningham virus) in individuals with impaired cell-mediated immunity.

ACTIVE LEARNING QUE

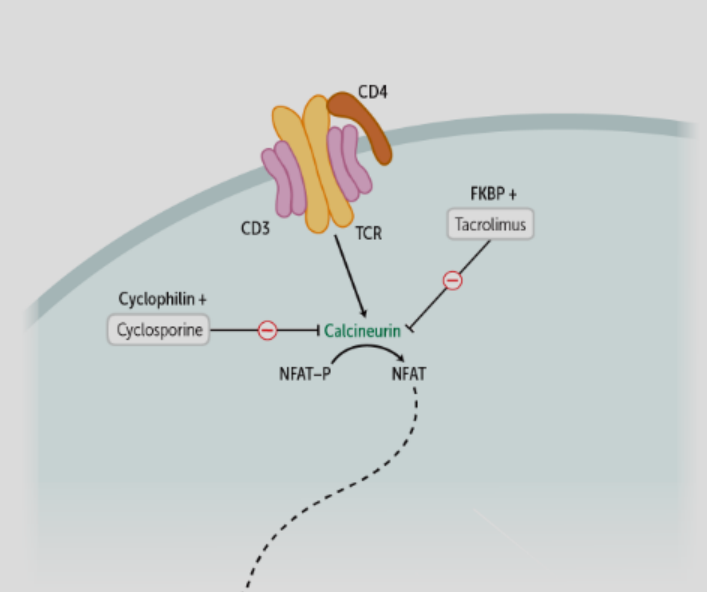
Vedolizumab reduces intestinal inflammation with minimal systemic immunosuppression. Which mechanism best explains this selectivity?

- **A.** ○ Systemic TNF-α neutralization
- **B.** ○ Gut-selective integrin blockade
- **C.** ○ Broad JAK pathway inhibition
- **D.** ○ Non-selective NF-κB suppression
- **E.** ○ Folate synthesis inhibition

Explanation (requires correct answer)

Cyclosporine

Mechanism of action: Cyclosporine is an inhibitor of calcineurin and a potent immunomodulator. Cyclosporine binds to cyclophilin which subsequently inhibits calcineurin. The inhibition of calcineurin inhibits the dephosphorylation of NF-AT (Nuclear Factor of Activated T cells). NF-AT activated genes include those required for B-cell activation including interleukin (IL)-4 and CD40 ligand and those required for T-cell activation including IL-2 and interferon gamma. It is effective in specific clinical settings in IBD, but the high f s limits its use as a first-line medication.



Adverse effects: The significant adverse effects associated with the use of cyclosporine limit its use to specific types of severe IBD. These side effects most often include increased susceptibility to infections, renal insufficiency, hypertension, seizures, and peripheral neuropathy. Significant drug interactions have been reported.

Monitoring strategies should be tailored to each drug's mechanism and expected toxicity profile. The table below summarizes key monitoring parameters for major

Table

Drug Class	Key Monitoring Parameter	Rationale
Aminosalicylates	Renal function (annual)	Risk of interstitial nephritis
Corticosteroids	Blood glucose, blood pressure, bone density	Metabolic and HPA axis effects
Thiopurines	CBC, LFTs, TPMT genotype	Myelosuppression and hepatotoxicity risk
Methotrexate	LFTs, CBC, pulmonary function	Hepatotoxicity and pneumonitis risk
Anti-TNF biologics	TB screening, hepatitis B serology	Risk of opportunistic infection reactivation

Why must TPMT activity be tested before starting thiopurine therapy in IBD?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MJ, what agents are most suitable for treating her inflammatory bowel disease, and how might their mechanisms explain her symptoms and potential side effects? MJ's bloody diarrhea and abdominal pain suggest active mucosal inflammation, consistent with moderate IBD. If her diagnosis is UC, a 5-ASA such as mesalamine would be appropriate first-line therapy, targeting mucosal NF-κB and COX pathways locally. If her disease is moderate-to-severe or involves CD, corticosteroids could induce remission, with transition to an immunomodulator or biologic for maintenance.

AJ should anticipate monitoring renal function if MJ is started on mesalamine, blood glucose and bone density if corticosteroids are used, and TPMT activity before initiating thiopurines. If a biologic is considered, TB and hepatitis B screening are mandatory. By linking each drug's mechanism to its expected side effects, AJ can ensure MJ receives effective therapy while minimizing her risk of preventable complications.

Summary

- Aminosalicylates (5-ASAs) act locally on the intestinal mucosa by inhibiting NF-κB, COX, and lipoxygenase pathways to reduce cytokine production and are first-line therapy for mild-to-moderate ulcerative colitis.
- Corticosteroids bind intracellular glucocorticoid receptors to broadly suppress inflammatory gene transcription and are effective for induction of remission but ance due to significant metabolic, endocrine, and systemic adverse effects.
- Immunomodulators, including thiopurines and methotrexate, reduce lymphocyte proliferation and cytokine production; thiopurines require TPMT genotyping before initiation to prevent life-threatening myelosuppression, and methotrexate requires folate supplementation to reduce mucosal toxicity.
- Biologic agents provide targeted intervention: anti-TNF antibodies (Infliximab, Adalimumab) neutralize TNF-α to suppress NF-κB signaling, vedolizumab selectively blocks integrin (α4β7) to limit gut-specific leukocyte trafficking, and ustekinumab inhibits IL-12/23 to suppress Th1/Th17-driven inflammation.
- Adverse effects are mechanistically predictable and guide monitoring: corticosteroids require glucose and bone density monitoring, thiopurines require CBC and LFTs, methotrexate requires pulmonary and liver function assessment, and anti-TNF biologics require screening for latent TB and hepatitis B prior to initiation.

Review Questions

1. Which adverse effect is commonly linked to long-term corticosteroid use in IBD?

- **A.** ○ Nephrotoxicity
- **B.** ○ Myelosuppression
- **C.** ○ Osteoporosis
- **D.** ○ Folate de
- **E.** ○ Pulmonary toxicity

Explanation (requires correct answer)

2. Why is TPMT testing required before starting thiopurine therapy?

- **A.** ○ To predict myelosuppression risk
- **B.** ○ To prevent corticosteroid toxicity
- **C.** ○ To avoid folate deficiency
- **D.** ○ To screen infection risk
- **E.** ○ To monitor renal toxicity

Explanation (requires correct answer)

3. A patient with Crohn's disease requires maintenance therapy. Which agent targets IL-12/23 to suppress Th1/Th17 inflammation?

- **A.** ○ Vedolizumab
- **B.** ○ Azathiopr
- **C.** ○ Ustekinumab
- **D.** ○ Infliximab
- **E.** ○ Budesonide

Explanation (requires correct answer)

4. A patient on infliximab develops fever and night sweats. Which pre-treatment screening was most critical?

- **A.** ○ TPMT genotyping
- **B.** ○ Latent TB screening
- **C.** ○ Bone density testing
- **D.** ○ Folate level testing

Explanation (requires correct answer)

5. A 29-year-old woman with moderate Crohn disease has persistent symptoms despite treatment with corticosteroids and thiopurines. Her gastroenterologist recommends switching to vedolizumab, noting that it is associated with n compared with other biologic agents. Which mechanism best explains the therapeutic effect of vedolizumab in inflammatory bowel disease?

- **A.** ○ Neutralization of soluble and membrane-bound tumor necrosis factor-α
- **B.** ○ Inhibition of Janus kinase–STAT signaling required for cytokine receptor activation
- **C.** ○ Blockade of α4β7 integrin–mediated lymphocyte trafficking to the intestinal mucosa
- **D.** ○ Suppression of NF-κB–dependent transcription of pro-inflammatory cytokines
- **E.** ○ Inhibition of purine synthesis leading to reduced T-lymphocyte proliferation

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Compare of drugs used to treat in

mucosa
- **D.** ○ Suppression of NF-κB–dependent transcription of pro-inflammatory cytokines
- **E.** ○ Inhibition of purine synthesis leading to reduced T-lymphocyte proliferation

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Compare and contrast the mechanisms of action of drugs used to treat inflammatory bowel disease
02 ☐ Analyze differences in mechanisms and adverse effects among inflammatory bowel disease treatment agents

Objective Confidence: 0/2